

DATASET INSPECTION REPORT

DATASET OVERVIEW

The Telco Customer Churn dataset contains 7,043 customer records and 21 variables. The dataset includes customer demographic information, service subscriptions, contract information, payment methods, billing information, tenure, and customer churn status.

The Churn variable represents the primary business outcome for this analysis and indicates whether a customer has left the service.

DATASET STRUCTURE

The dataset contains a combination of numerical and categorical variables.

Numerical variables include SeniorCitizen, tenure, and MonthlyCharges, while categorical variables include customer demographics, subscribed services, contract type, payment method, and churn status.

The customerID field serves as a customer identifier.

DATA TYPES

The TotalCharges variable has a String data type instead of a numeric data type (int) as expected. After investigation, it was concluded that this is due to the presence of 11 blank spaces in the column

MISSING AND BLANK VALUES

A standard null-value check did not identify any NaN values. However, further inspection of blank strings identified 11 blank values in the TotalCharges field.

Investigation showed that these records correspond to customers with zero months of tenure.

DUPLICATE RECORDS

The dataset contains no duplicate rows.

DESCRIPTIVE STATISTICS

The average customer tenure is approximately 32.37 months, with tenure ranging from 0 to 72 months.

The average monthly charge is approximately 64.76, with observed monthly charges ranging from 18.25 to 118.75.

INITIAL DATA QUALITY ASSESSMENT

Overall, the dataset is suitable for further analysis. But, the following data-quality issues must be addressed before proceeding:

- TotalCharges is stored as a string/object variable rather than a numeric variable.
- 11 records contain blank TotalCharges values.
- The blank records are associated with customers whose tenure is zero.

No duplicate rows or duplicate customer IDs were identified.

The dataset will therefore proceed to the data-cleaning stage, where data types and blank values will be appropriately handled before exploratory and business analysis.